

ULIP Integration Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	09/01/2025	Ayush Arya	Nayab Ali	Baseline Version
2.0	08/01/2026	Hasan Mhad Khan	Nayab Ali	Payload Update for IOCL/02
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for IOCL system information (information about Indian oil and natural gas stations based on stateCode or geographical location (longitude, latitude, and radius) value from ULIP system

This document details the following points:

1. **Data Integration between IOCL and ULIP web service:** Identification of operational points where data integration between **IOCL** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will need to share their stateCode or geographical location (longitude, latitude, and radius) to access the list of public petroleum and natural gas stations through the rest API.

Here are some points to keep in mind when accessing IOCL API data information:

- The API will only share data that is based on the user's provided information.
- ULIP will expose API for getting data from IOCL API.
- The IOCL/01 API will take user information's from user for register that user. And user will receive ReferenceNo in response, from this ReferenceNo user will be able to access the IOCL/02 API.
- The IOCL/02 API will take ReferenceNo and stateCode or geographical location (longitude, latitude, and radius) to give the information about petroleum and natural gas stations to the user.

1.2 Data Integration between IOCL System and ULIP application

The ULIP system will receive information from the user and provide data in return.

1.3 IOCL/01

This API is designed to register the user to the IOCL system based on user information. User will provide the user information and once user successfully register, then they will get the reference number in the response.

Please note that when making a request, **all of the JSON keys are mandatory in the payload**

```
{
  "customerName": "Ayush",
  "address": "Gautam Bhuda",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
  "userIP": "193.2.2.2",
  "mobileNo": "9161666858"
}
```

	Field Name	Field Description	Format	Required/Not – Required	Length
1	customerName	Customer Name	^[a-zA-Z\s]{2,20}\$	Required	2-20
2	address	Address	^[a-zA-Z][a-zA-Z0-9\s,.-]{1,49}\$	Required	2-49
3	district	District	^[a-zA-Z\s]{2,20}\$	Required	2-20
4	state	State	^[a-zA-Z\s]{1,30}\$	Required	1-30
5	email	Email	[A-Za-z0-9\._%+\-]+@[A-Za-z0-9\.\-]+\.[A-Za-z]{2,}	Required	
6	monthlyFuelConsumption	Monthly fuel consumption	^[0-9]{1,10}\$	Required	1-10
7	userId	User ID	^[a-zA-Z][a-zA-Z0-9\s._@]{1,20}\$	Required	2-20
8	userIP	User IP	^(25[0-5] 2[0-4][0-9] [01]?[0-9][0-9]?)\.(25[0-5] 2[0-4][0-9] [01]?[0-9][0-9]?)\.(25[0-5] 2[0-4][0-9] [01]?[0-9][0-9]?)\.(25[0-5] 2[0-4][0-9] [01]?[0-9][0-9]?)\$	Required	
9	mobileNo	Mobile number	^[0-9]\d{9}\$	Required	10

1.3.1 Technical Approach

1. All the data will be shared through REST web services.
2. The user will share the input for which information is required.
3. IOCL API will share details of IOCL API information that exists in the system in JSON format.
4. The ULIP system will provide data in JSON Object format in the response body.
5. When we hit the IOCP/01 and the user successfully registers, a reference number is returned.

Request:

The request must be of the following format: **user information** is passed within the URL post request body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/01>

```
{
  "customerName": "Ayush",
  "address": "Gautam Bhuda",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
  "userIP": "193.2.2.2",
  "mobileNo": "9161666858"
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWUiOiJ1bGlwliwiaWF0IjoxNzY2MzY2NzgwLCJhcHBzIjoIZGF0YXB1c2gifQ.kV2pvPVIZtrXJt_1crjumGutjk_WloCo-xPI8Vmx9vBuR0v3GPpWL11nSFFU60GprznpcgkiOQ-eNyePh_BwA' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{
  "customerName": "Ayush",
  "address": "Gautam Bhuda",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
}
```

```
"userIP": "193.2.2.2",  
"mobileNo": "9161666858"  
}
```

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/IOCL/01>

```
{  
  "customerName": "Ayush",  
  "address": "Gautam Bhuda",  
  "district": "Gautam Bhuda Nagar",  
  "state": "Uttar Pradesh",  
  "email": "test@paytm.com",  
  "monthlyFuelConsumption": "10000",  
  "userId": "test",  
  "userIP": "193.2.2.2",  
  "mobileNo": "9161666858"  
}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/IOCL/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzMzMzk2NzgwLCJhcHBzIjoZGF  
0YXB1c2gifQ.kV2pvPVIZtrxJt_1crjumGutjk_WloCo-  
xPI8Vmx9vBuR0v3GPpWL11nSFFU60GprznpckgiOQ-eNyePh_BwA' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{  
  "customerName": "Ayush",  
  "address": "Gautam Bhuda",  
  "district": "Gautam Bhuda Nagar",  
  "state": "Uttar Pradesh",  
  "email": "test@paytm.com",  
  "monthlyFuelConsumption": "10000",  
  "userId": "test",  
  "userIP": "193.2.2.2",  
  "mobileNo": "9161666858"  
}'
```

For obtaining an authentication token, refer to [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the response below -

a) In case of an invalid format**1. Invalid email**

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: email. Format should follow [A-Za-z0-9\\._%+\\-]+@[A-Za-z0-9\\.\\-]+\\. [A-Za-z]{2,}"
}
```

2. Invalid userIP format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: userIP. Format should follow ^(((25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?\\.\\.){3}(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?\\.)*((25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?\\.\\.){3}(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?$)"
}
```

3. Invalid mobile format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: mobileNo. Format should follow ^[0-9]\\d{9}$"
}
```

4. In case IOCL API data user information already exists in the IOCL system

```
{
  "response": [
    {
      "response": {
        "ReferenceNo": null,
        "StatusCode": 1,
        "Message": "mobileNo already exists."
      },
      "responseStatus": "ERROR"
    }
  ],
  "error": "false",
  "code": "200",
}
```

```

"message": "Success"
}

```

5. In case IOCL API data user information does not exist in the IOCL system

```

{
  "response": {
    "response": {
      "ReferenceNo": null,
      "StatusCode": 1,
      "Message": "mobileNo already exists."
    },
    "responseStatus": "ERROR"
  },
  "error": "false",
  "code": "200 OK",
  "message": "Success"
}

```

Status Code 200: IOCL API data response

Status Code 400: Bad request (Invalid JSON syntax, invalid JSON key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: The server is not responding.

1.4 IOCL/02

This API is designed to retrieve information about petroleum and natural gas stations. It can be used by providing either the stateCode or geographical location (longitude, latitude, and radius) of the desired location.

IOCL merchant(outlet) details are updated weekly, and retail selling prices (RSP) are updated daily to ensure users always see the latest and accurate information.

Use the state code from the table below to inquire about the HPCL outlet in the respective state.

Serial no.	State Code	State Name
1	AN	Andaman & Nicobar
2	AP	Andhra Pradesh
3	ARP	Arunachal Pradesh
4	AS	Assam
5	BH	Bihar
6	CD	Chandigarh
7	CSG	Chhatisgarh
8	DDH	UT OF D&NH AND D&D

9	DDH	UT OF D&NH AND D&D
10	DEL	Delhi
11	GDD	Goa
12	GJ	Gujarat
13	HP	Himachal Pradesh
14	HR	Haryana
15	JK	Jammu & Kashmir
16	JRK	Jharkhand
17	KAR	Karnataka
18	KER	Kerala
19	LAK	Lakshwadeep
20	LDK	Ladakh
21	MAH	Maharashtra
22	MGL	Meghalaya
23	MNP	Manipur
24	MP	Madhya Pradesh
25	MZ	Mizoram
26	NZ	Nagaland
27	OR	Odisha
28	PB	Punjab
29	PY	Pondicherry
30	RJ	Rajasthan
31	SK	Sikkim
32	TG	Telangana
33	TN	Tamil Nadu
34	TRP	Tripura
35	UP	Uttar Pradesh
36	UTK	Uttarakhand
37	WB	West Bengal

Please note that when making a request, **all of the JSON keys are mandatory in the payload**. First, the user must enter the reference number. If this reference number exists on the IOCL side, the user can then enter either the stateCode or geographical location (longitude, latitude, and radius). The reference number can be obtained when they hit the IOCL/01 API.

If you choose to use the stateCode to access IOCL API data, then geographical location (longitude, latitude, and radius) fields should be left empty.

```
{
  "referenceNo": "9161666748",
  "stateCode": "WB",
```

```
{
  "longitude": "",
  "latitude": "",
  "radius": ""
}
```

If you choose to use the geographical location to access IOCL API data, then stateCode field should be left empty.

```
{
  "referenceNo": "7878942650",
  "stateId": "",
  "longitude": "74.44155",
  "latitude": "20.25412",
  "radius": "2"
}
```

	Field Name	Field Description	Format	Required/Not – Required	Length
1	referenceNo	Reference number	^[0-9]\d{9}\$	Required	10
2	stateCode	State ID	^[A-Z]{1,6}\$	Required	1-6
4	longitude	Longitude	^[0-9]{2}\.{0,1}[0-9]{2,15}[0-9]{0,2}\$	Required	0-18
5	latitude	Latitude	^[0-9]{2}\.{0,1}[0-9]{2,15}[0-9]{0,2}\$	Required	0-18
6	radius	Radius	^\d{0,3}(\.\d{1,10})?\$	Required	1-10

1.4.1 Technical Approach

1. All the data will be shared through REST web services.
2. The user will share the input for which information is required.
3. IOCL API will share details of IOCL API station information that exists in the system in JSON format.
4. The ULIP system will provide data in JSON Object format in the response body.

Request:

The request must be of the following format, where **stateCode** or **geographical location (longitude, latitude, and radius)** are passed within the URL post request body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/02>

```
{
  "referenceNo": "9161666748",
  "stateCode": "WB",
  "longitude": "",
  "latitude": "",
  "radius": ""
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/02' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzY2MzY2MzgwLCJhcHBzIjoZGF0YXB1c2gifQ.kV2pvPVIztrrxJt_1crjumGutjk_WloCo-xPI8Vmx9vBuR0v3GPpWL11nSFFU60GprznpcgkiOQ-eNyePh_BwA' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "referenceNo": "9161666748",
  "stateCode": "WB",
  "longitude": "",
  "latitude": "",
  "radius": ""
}'
```

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/IOCL/02>

```
{
  "referenceNo": "9161666748",
  "stateCode": "WB",
  "longitude": "",
  "latitude": "",
  "radius": ""
}
```


3. Invalid radius format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: radius. Format should follow ^\\d{0,3}(\\.\\d{1,10})?$"
}
```

4. Invalid stateCode format

```
{
  "response": null,
  "error": "true",
  "code": "400 BAD_REQUEST",
  "message": "Data format failed OR wrong value entered at: stateCode. Format should follow ^[A-Z]{1,6}$"
}
```

5. In case IOCL API referenceNo does not exist in the IOCL system

```
{
  "response": {
    "response": {
      "errorMsg": "referenceNo is not valid",
      "errorCode": "PAYLOAD_ERROR"
    },
    "responseStatus": "ERROR"
  },
  "error": "false",
  "code": "200 OK",
  "message": "Success"
}
```

6. In case IOCL API data exist

```
{
  "response": {
    "response": [
      {
        "CUSTOMER_CODE": 371451,
        "CUSTOMER_NAME": "ADHOC-SWARNALATA PETRO SEVA",
        "ADDRESS": "C/O- HALDAR SERVICE STATION,KHOLAPOTA TAKI ROAD,BASIRHAT",
        "PIN_CODE": 743428,
        "LONGITUDE": 88.814714,
        "LATITUDE": 22.682907,

```

```
"STATE_CODE": "WB",
"STATE_NAME": "West Bengal",
"DISTRICT_CODE": "WB015",
"DISTRICT_NAME": "NORTH 24 PARGANAS",
"PRODUCT_NAMES": "Diesel,Petrol",
"FACILITIES": "FREE AIR FACILITY,CLEAN DRINKING
WATER,WASHROOM,CREDIT CARD,AUTOMATED
OUTLET,XTRAPOWER,XTRAREWARD,RESTROOM FOR MEN AND WOMAN",
"PRODUCT_DETAILS": [
  {
    "EFFECTIVE_DATE": "08-01-2025",
    "PRODUCT_NAME": "HSD",
    "PRICE": 92.3
  },
  {
    "EFFECTIVE_DATE": "08-01-2025",
    "PRODUCT_NAME": "MS",
    "PRICE": 105.53
  }
]
},
{
  "CUSTOMER_CODE": 372542,
  "CUSTOMER_NAME": "Jirat Fuel Station",
  "ADDRESS": "INDIAN OIL RETAIL OUTLET,Plot No. 828 Mouza & Vill-
Mundukh,Hooghly",
  "PIN_CODE": 712501,
  "LONGITUDE": null,
  "LATITUDE": null,
  "STATE_CODE": "WB",
  "STATE_NAME": "West Bengal",
  "DISTRICT_CODE": "WB008",
  "DISTRICT_NAME": "HOOGHLY",
  "PRODUCT_NAMES": null,
  "FACILITIES": "FREE AIR FACILITY,CLEAN DRINKING WATER,WASHROOM",
  "PRODUCT_DETAILS": [
    {
      "EFFECTIVE_DATE": "08-01-2025",
      "PRODUCT_NAME": "HSD",
      "PRICE": 92.3
    },
    {
      "EFFECTIVE_DATE": "08-01-2025",
      "PRODUCT_NAME": "MS",
      "PRICE": 105.53
    }
  ]
},
{
```

```

    "CUSTOMER_CODE": 220851,
    "CUSTOMER_NAME": "BSS KISAN SEVA KENDRA",
    "ADDRESS": "VILL: BARDANGA,POST: SARAI DANGA PS:
GAZOLE,BARDANGA",
    "PIN_CODE": 732124,
    "LONGITUDE": null,
    "LATITUDE": null,
    "STATE_CODE": "WB",
    "STATE_NAME": "West Bengal",
    "DISTRICT_CODE": "WB011",
    "DISTRICT_NAME": "MALDA",
    "PRODUCT_NAMES": null,
    "FACILITIES": "FREE AIR FACILITY,CLEAN DRINKING WATER,WASHROOM",
    "PRODUCT_DETAILS": []
  },
  {
    "CUSTOMER_CODE": 366313,
    "CUSTOMER_NAME": "ADHOC SHIB NARAYAN FILLING STATION",
    "ADDRESS": "C/O KIRISTASWARI OIL TRADERS IOCL,VILL: BRAMHANI GRAM
PS: SAGARDIGHI,BRAHMANI GRAM",
    "PIN_CODE": 742226,
    "LONGITUDE": null,
    "LATITUDE": null,
    "STATE_CODE": "WB",
    "STATE_NAME": "West Bengal",
    "DISTRICT_CODE": "WB013",
    "DISTRICT_NAME": "MURSHIDABAD",
    "PRODUCT_NAMES": null,
    "FACILITIES": "FREE AIR FACILITY,CLEAN DRINKING WATER,WASHROOM",
    "PRODUCT_DETAILS": []
  }
],
"responseStatus": "SUCCESS"
},
"error": "false",
"code": "200 OK",
"message": "Success"
}

```

7. In case IOCL API data do not exist

```

{
  "response": {
    "response": [],
    "responseStatus": "SUCCESS"
  },
  "error": "false",
  "code": "200 OK",

```

```
"message": "Success"
}
```

Status Code 200: IOCL API data response

Status Code 400: Bad request (Invalid JSON syntax, invalid JSON key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: The server is not responding.

1.5 Data Transmission & Authentication Mechanism

All data exchanges will be securely done with HTTPS. ULIP's system will request data from IOCL through a URL with the following format:

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/01>

Body:

```
{
  "customerName": "test",
  "address": "test address",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
  "userIP": "193.2.2.2",
  "mobileNo": "7879877897"
}
```

To access the data exchange URL, you would need to use a security mechanism that is implemented by the ULIP system. The process is as follows:

- First time access: you will require a username and password. The username and password would be shared with you beforehand. Once you enter the valid username and password, the ULIP system will return a basic authorization token to you. You can use this token to communicate with the ULIP system until the token gets expired.

In brief, HTTP Basic Authentication works as follows:

When you hit the ULIP secured API for the first time, you need to provide a username and password to get an access token.

```
curl --location
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
```

```
--header 'Content-Type: application/json' \
--data-raw '{
  "username": "xxxx",
  "password": "xxxx"
}'
```

Authorization:**Bearer**

```
eyJshbGciqijlazmxMiJ8.eyJzdWliOiJ1bGlwliwiaWF0ljoXNzI2NTU2ODk4LCJhcHBzljoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-9TWWpQQLUX6CsUO5drkwmny9hU4hl6cOr8CXHIJ67Nv-HdGNtHtr2bH8Irg
```

After successfully authenticating, the user will be logged into the ULIP system and obtain an access token. This token will be used by the user to communicate with the ULIP system. The token has an expiry time that represents the user session, which is typically set to 30 minutes. If there is no request from the user within this time frame, the session (token) will expire, and the user will need to re-authenticate.

Using The above token, user can request ULIP system for IOCL API for details as follow -

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/IOCL/01' \
--header 'Authorization: Bearer eyshbGciqijlazmxMiJ8.eyJzdWliOiJ1bGlwliwiaWF0ljoXNzI2NTU2ODk4LCJhcHBzljoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-9TWWpQQLUX6CsUO5drkwmny9hU4hl6cOr8CXHIJ67Nv-HdGNtHtr2bH8Irg' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{
  "customerName": "test",
  "address": "S.",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
  "userIP": "193.2.2.2",
  "mobileNo": "7878992650"
}'
```

For production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{
  "username": "xxxx",
```

```
"password": "xxxx"
}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzI1NiIsInR5cGEiOiJ1bGlwliwiaWF0IjoxNzI2NTU2ODk4LCJhcHBzIjoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-9TWWpQQLUX6CsUO5drkwmny9hU4hlf6cOr8CXHIJ67Nv-HdGNtHtr2bH8lrg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/IOCL/01>

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/IOCL/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cGEiOiJ1bGlwliwiaWF0IjoxNzI2NTU2ODk4LCJhcHBzIjoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-9TWWpQQLUX6CsUO5drkwmny9hU4hlf6cOr8CXHIJ67Nv-HdGNtHtr2bH8lrg
\
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
  "customerName": "test",
  "address": "S.",
  "district": "Gautam Bhuda Nagar",
  "state": "Uttar Pradesh",
  "email": "test@paytm.com",
  "monthlyFuelConsumption": "10000",
  "userId": "test",
  "userIP": "193.2.2.2",
  "mobileNo": "7878192650"
}'
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